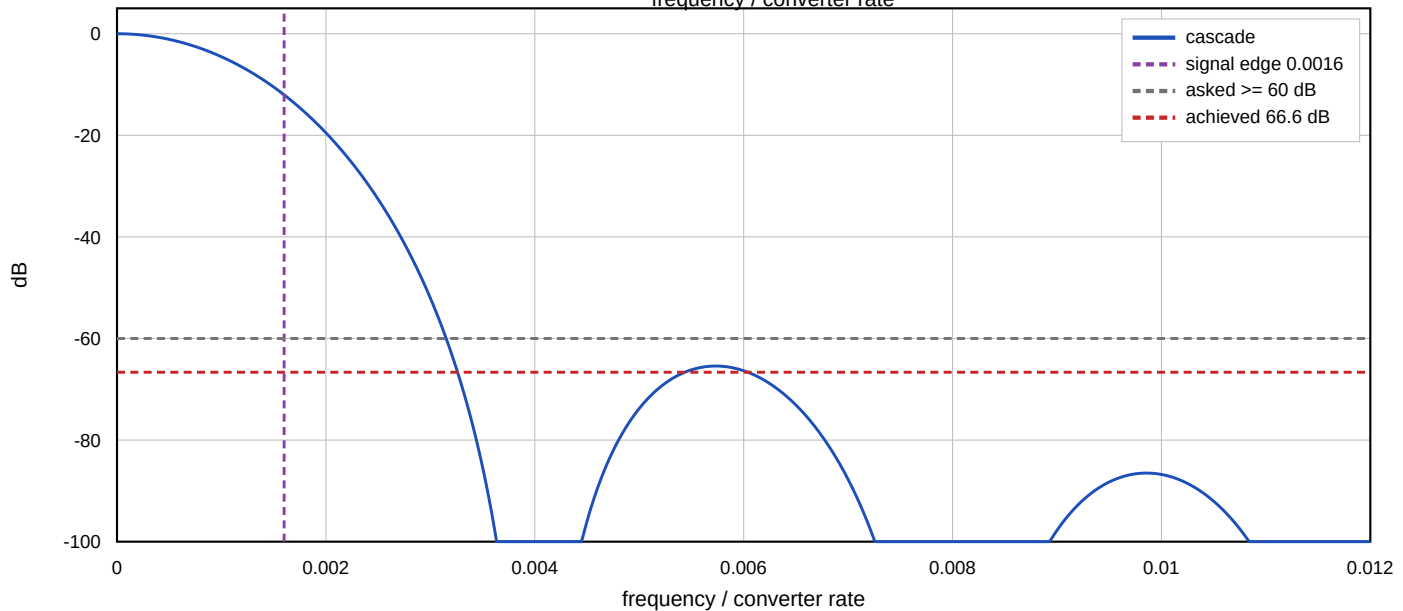
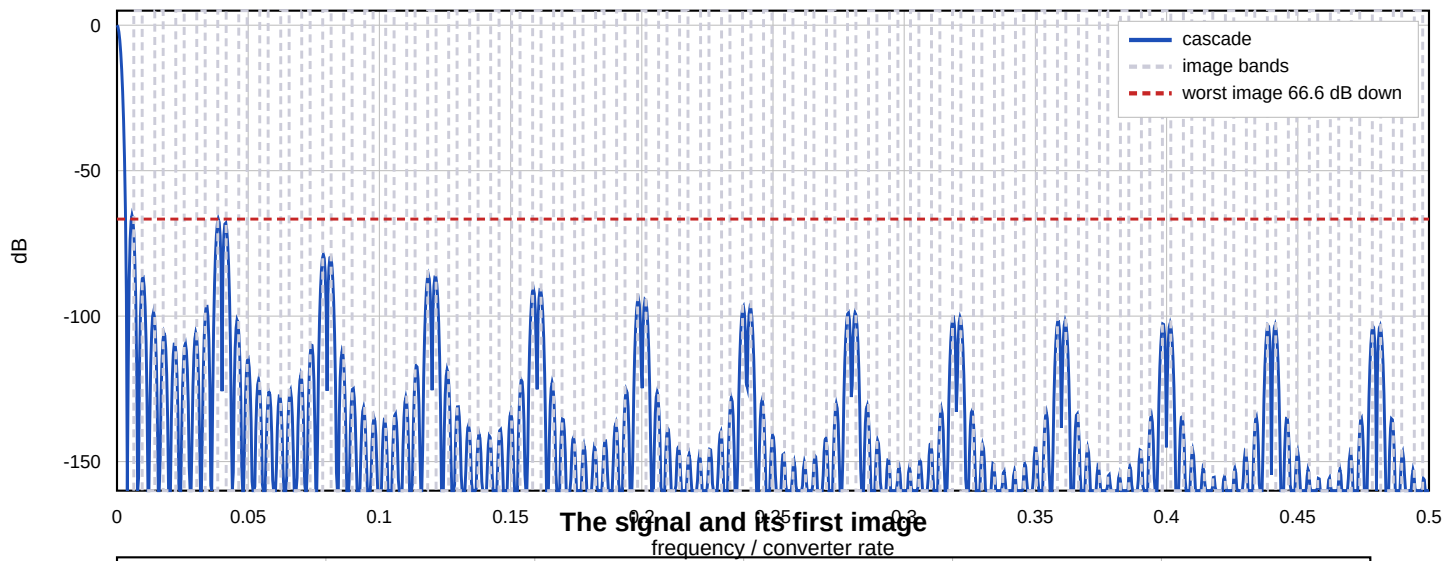


Interpolation Chain - Derived Design

1000.0000 kHz x 125 = 125.000 MHz, split [5, 25], 200.000 kHz signal

Composite magnitude across the converter band



Stages

stage 1: x5 N=5 M=2 combs at 1.0000 MHz, integrators at 5.000 MHz

in 16 bits, uniform accumulator 31 bits, tapered widths [17, 18, 19, 20, 21, 21, 24, 26, 28, 31]

620 register bits uniform, 415 as built (lossless), bound 415

widths as built [17, 18, 19, 20, 21, 21, 24, 26, 28, 31]

stage 2: x25 N=2 M=1 combs at 5.0000 MHz, integrators at 125.000 MHz

in 31 bits, uniform accumulator 36 bits, tapered widths [32, 33, 32, 36]

216 register bits uniform, 199 as built (lossless), bound 198

widths as built [32, 33, 33, 36]

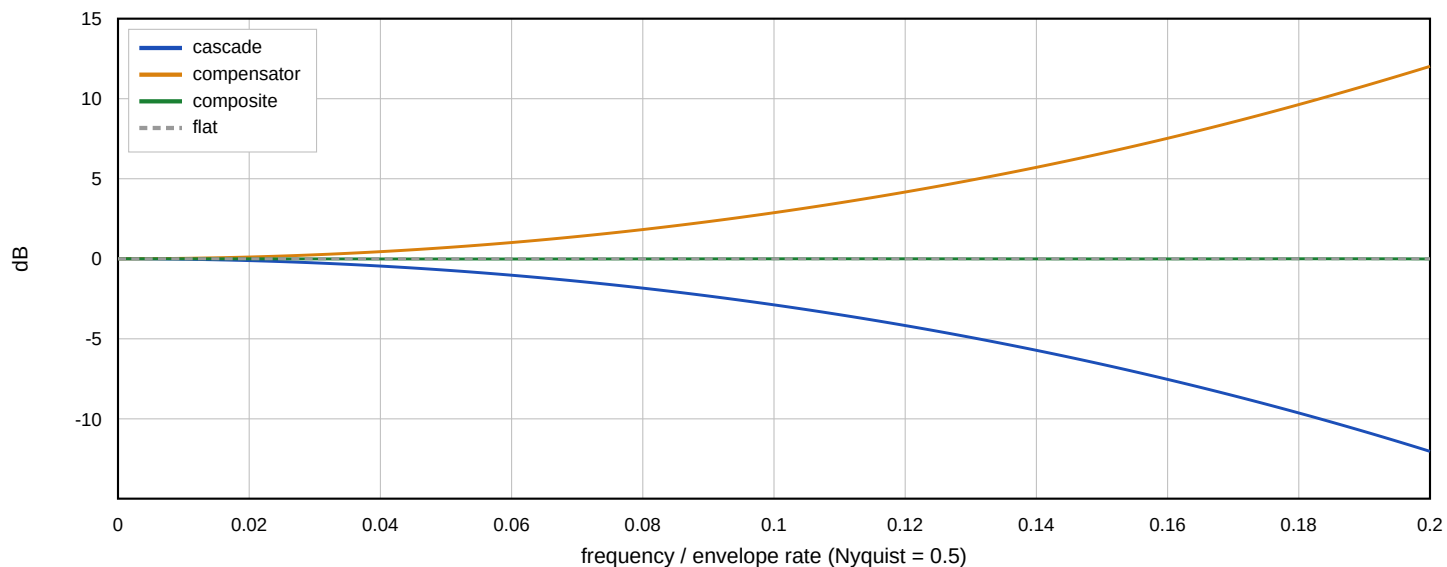
The combs run at the stage's input rate and the integrators at its output rate, so an integrator bit costs R times what a comb bit costs. Depth belongs early and rate late.

The tapered widths are lossless: each stage sized to its own growth bound holds its value exactly, so a tapered interpolator is bit-identical to a uniform one.

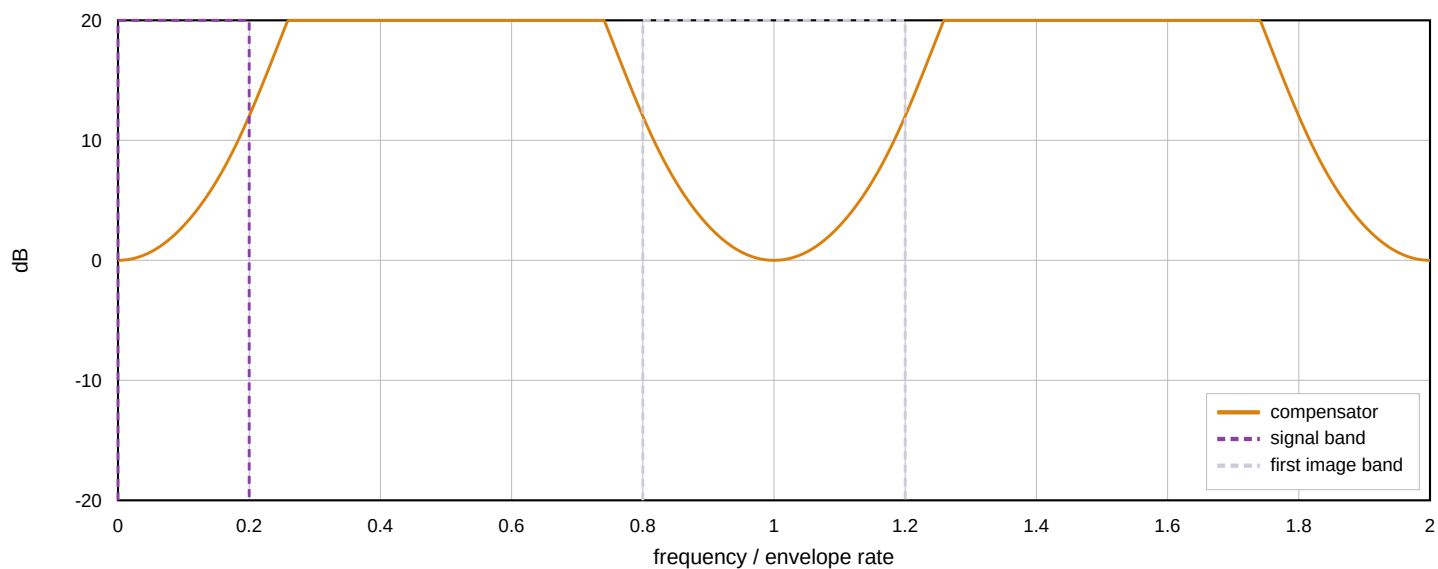
Compensation and Result

9 taps, 16-bit coefficients, at the 1.0000 MHz envelope rate

Droop, pre-compensator, and the two together - the signal band



The compensator repeats, so it lifts every image as much as the signal



Figures and Cautions

Figures

images 66.6 dB down (asked ≥ 60.0)
 worst image at 4.8297 MHz
passband ripple 0.0147 dB (asked ≤ 0.100)
uncompensated droop .. -11.986 dB at the band edge
converter floor 86.0 dB at 14 bits (the chain adds none)
register bits 836 uniform, 614 as built, 613 at the exact bound
compensator DC gain .. 1.0000, shift 10
compensator headroom . 23 bits for any input, 18 for in-band only
 norms l1 81.1289, passband peak 3.9907
rates reachable 73 of 124 in $2..=125$ (a split restricts them)
worst image at rate .. 125
adder depth 1 in the combs, 1 in the integrators (both pipelined)

group delay 1864 converter samples = 14.9 μ s
 cascade 586, integrator pipeline 101, comb pipeline 675, output regs 2, compensator 500
 largest term is the comb pipeline at 675 (36%); loop bandwidth ~ 6.7 kHz
runner-up split [5, 25] N=[5, 2] M=[2, 2], 924 register bits

taps [304, -2290, 8288, -18223, 24866, -18223, 8288, -2290, 304]

Build to the any-input headroom unless the envelope is band-limited.

The in-band figure is the peak gain an in-band sinusoid sees. The any-input figure is the sum of the taps' magnitudes, which is the bound for an arbitrary bounded input -- and a transmit envelope is not band-limited: switch-on, a burst boundary and a modulation change are all steps, and a step is exactly the input that reaches it. Choosing the narrower figure for a burst transmitter is how a compensator saturates on the first sample.

More taps will not improve image rejection.

The compensator runs before the rate change, at the envelope rate, so its response is periodic and the image at $k+u$ sees exactly the gain the signal at u sees -- the plot above shows it. The image-to-signal ratio is the cascade's alone. This is the one place where transposing receive-side intuition is wrong: there, the compensator sits after the fold and its stopband is part of the alias budget.

The knobs are the CIC depth N , the interpolation factor R , and the signal bandwidth. A compensator at the converter rate would work, because it is no longer periodic in u , and costs a FIR running at the full clock.